**Sharif University of Technology**

Department of Electrical Engineering

# EMG and Strain-Gauge Signal Conditioning

A Dual-Channel Analog Measurement System

**Sensing and Measurement — Course Project Report**

Sanaz Mazrae

Instructor: Dr. Jahed | February 2025

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## 1. Project Objective and Scope

The project addresses the analog acquisition of two low-level signals associated with a forearm movement performed while holding and lowering a load: surface electromyography (sEMG) from the triceps brachii and the electrical output of a strain-gauge sensing element at the grip. The aim is to develop signal-conditioning circuits that reject unwanted components, amplify the differential sensor signals, and provide voltage levels suitable for subsequent acquisition.

Two separate signal paths were designed and simulated in **Proteus**; a related implementation, including an exploratory analog-to-digital conversion stage, was modeled in **LTspice**. The submitted material contains schematic files and simulated waveforms, but no physical measurements, electrode recordings, measured forces, or calibrated strain-gauge data. Accordingly, this report evaluates the circuit designs and their simulated behavior, not experimental recovery of muscle force or forearm kinematics.

The principal engineering tasks are differential signal acquisition, frequency-selective filtering, gain allocation, mitigation of 50-Hz interference, and DC-offset management. The analog front ends are considered independently because the source amplitudes and frequency content differ substantially.

## 2. Signal Characteristics and Design Requirements

**Table 1:** Signal assumptions used in the project design.

| | **Surface EMG** | **Strain-gauge output** |
|---|---|---|
| Nominal signal scale | Approximately 100 $\mu$V | Approximately 1 mV |
| Frequency range discussed in the design | 20–500 Hz | Approximately 1–100 Hz |
| Measurement configuration | Differential surface-electrode signals | Differential voltage used as a proxy for bridge/sensor output |
| Dominant conditioning needs | High common-mode rejection; rejection of motion, high-frequency and mains interference | Low-frequency preservation; amplification; rejection of high-frequency components and offsets |

The signal scales are design assumptions rather than amplitudes obtained from a recorded experiment. Similarly, the simulated strain signal represents a sensor output voltage; the supplied files do not establish a mechanical force-to-voltage calibration, gauge factor, Wheatstone-bridge sensitivity, or measurement uncertainty. No numerical force estimate is therefore reported.

A front-end gain must be large enough to make these small voltages observable without saturating the amplifiers. Filtering and gain cannot be selected independently: the passband determines which physiological or mechanical components survive, while gain and common-mode rejection determine the usable dynamic range and interference sensitivity.

# 3. System Architecture

The two conditioning chains follow the same broad strategy: suppress unwanted frequencies, buffer the signal where necessary to reduce interstage loading, amplify the differential component, and remove the remaining DC component before interfacing with a downstream acquisition circuit.

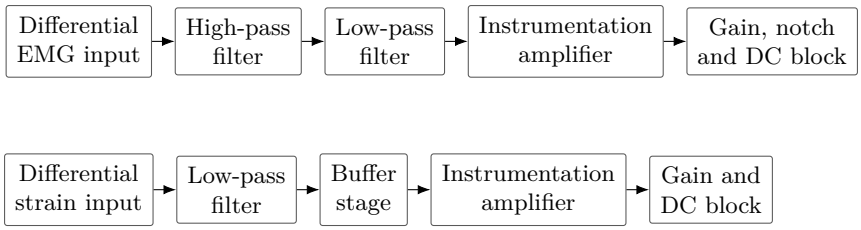



**Figure 1:** Functional signal paths used to organize the Proteus design. Buffers are present between filtering stages in the implemented schematic; the diagram condenses repeated buffers for readability.

Both paths use instrumentation amplification to detect a small differential voltage in the presence of common-mode interference. The EMG chain additionally includes a 50-Hz notch stage. This distinction follows the original design choice to give the strain-gauge channel a relatively low upper cutoff instead of introducing a separate notch section.

# 4. Proteus Circuit Design

## 4.1 EMG signal-conditioning path

The EMG front end uses high-pass and low-pass Sallen–Key sections, buffers, an instrumentation amplifier, a non-inverting gain stage, a 50-Hz notch filter, and a final DC-blocking network. The high-pass section is intended to reduce slowly varying offsets and motion-related low-frequency components, while the low-pass section limits the high-frequency content reaching the gain stages.

The original design calculations give nominal high-pass and low-pass corner frequencies of approximately 20 Hz and 200 Hz. For the component arrangements indicated in the design, the nominal frequency relations are

$$f_{\mathrm{HP}} \simeq \frac{1}{2\pi\sqrt{R_3 R_4 C_4 C_5}} \approx 20\,\mathrm{Hz}, \qquad f_{\mathrm{LP}} \simeq \frac{1}{2\pi\sqrt{R_{14} R_{15} C_3 C_6}} \approx 200\,\mathrm{Hz}. \tag{1}$$

These expressions specify characteristic frequencies. A complete second-order frequency response also depends on the damping factor, stage gain, tolerances, and the loading presented by the adjoining blocks. In particular, choosing a 200 Hz upper corner attenuates part of the broader 20–500 Hz EMG range identified in the introductory requirements.

The instrumentation and non-inverting stages have the nominal voltage gains

$$A_{\mathrm{IA,EMG}} = \left(1 + \frac{2R_{28}}{R_{25}}\right)\frac{R_{24}}{R_{22}} = 164, \qquad A_{\mathrm{NI,EMG}} = 1 + \frac{R_{21}}{R_{27}} = 41, \tag{2}$$

which implies a nominal cascaded midband gain of

$$A_{\mathrm{EMG,nominal}} = 164 \times 41 = 6724 \text{ V/V} \quad (\approx 76.6\,\mathrm{dB}). \tag{3}$$

This is the product of ideal stage gains; it is not an independently measured overall gain. Filter attenuation, the notch response, headroom, and the applied differential input determine the output of the complete chain. A 100 $\mu$V differential component would correspond to about 0.67 V before filter attenuation if all stages were linear and at their nominal gains.

To suppress power-line coupling, the EMG design incorporates a notch centered nominally at 50 Hz. Its characteristic-frequency calculation was given as $f_0 = 1/(4\pi R_{36}C_{18}) = 50$ Hz for the specific circuit arrangement. Since 50 Hz also lies inside the physiological EMG band, notch filtering necessarily removes any desired signal component near that frequency. A DC-blocking stage follows the signal-processing chain to reduce residual offsets.

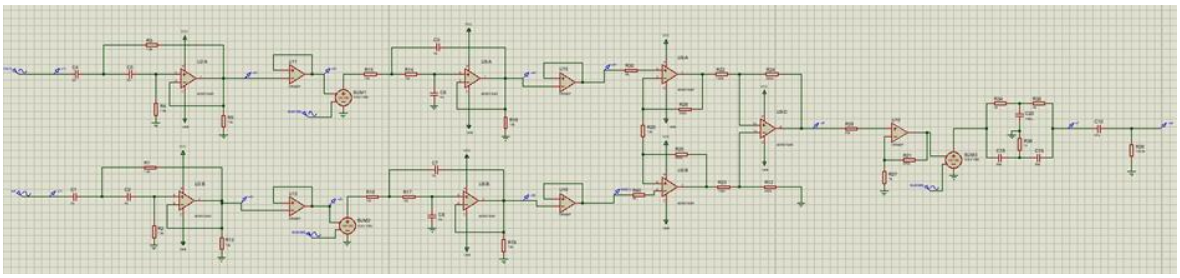


**Figure 2:** Complete Proteus EMG conditioning circuit, showing the dual input paths, differential gain, notch stage, and output network.

## 4.2 Strain-gauge signal-conditioning path

The strain-gauge path uses a low-pass filter, buffering, instrumentation amplification, a non-inverting amplifier, and a DC-blocking output section. The design does not implement a calibrated force measurement; it conditions a simulated differential voltage at the level expected from the sensor.

For the first-order low-pass section, the recorded design relation is

$$f_{\mathrm{LP,strain}} = \frac{1}{2\pi R_{33}C_{11}} \approx 35\,\mathrm{Hz}. \tag{4}$$

The design text also describes a nominal target near 30 Hz. This is suitable for slowly varying load signals, but it will attenuate faster variations in the originally stated 1–100 Hz range. The instrumentation and non-inverting gain expressions are

$$A_{\mathrm{IA,strain}} = \left(1 + \frac{2R_{31}}{R_{11}}\right)\frac{R_{10}}{R_8} = 20, \qquad A_{\mathrm{NI,strain}} = 1 + \frac{R_{38}}{R_{39}} = 11, \tag{5}$$

so that the calculated midband gain is

$$A_{\mathrm{strain,nominal}} = 20 \times 11 = 220 \text{ V/V} \quad (\approx 46.8\,\mathrm{dB}). \tag{6}$$

For a 1 mV differential input this gain corresponds to 0.22 V before the effects of the filters. The output DC-blocking network is a high-pass stage with a small but nonzero cutoff; an exactly

zero high-pass cutoff would not remove DC.

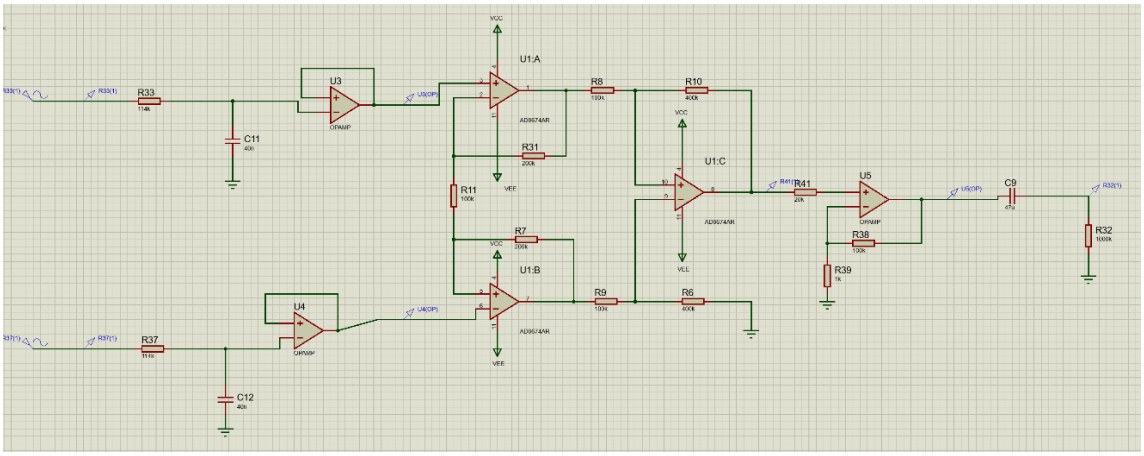



**Figure 3:** Complete Proteus strain-gauge conditioning circuit. The sensor is represented by a differential electrical input rather than a physical bridge and mechanical load model.

# 5. Simulation Conditions and Supplementary LTspice Model

## 5.1 Proteus test methodology

The Proteus simulations use sinusoidal signals at representative frequencies, combined with interfering components to inspect the behavior at intermediate nodes. Differential sources were used for the strain-gauge input, and waveforms were inspected before and after filters, buffers, gain stages, and output coupling. These tests establish qualitative stage behavior and amplitude changes, but the submitted report does not contain an automated sweep of gain, phase, input-referred noise, or CMRR.

## 5.2 LTspice circuit configuration

A separate LTspice schematic, `Project.asc`, contains analog filters, OP07 buffer/amplifier symbols, AD8429 instrumentation amplifiers, behavioral noise sources, and two optional ADC/isolation branches. It is not component-for-component identical to the Proteus design, so its component values must be interpreted as a separate implementation.

**Table 2:** Settings readable directly from the supplied LTspice schematic.

| Item | Saved setting |
|---|---|
| EMG signal sources | Two 150 Hz sinusoids, 100 $\mu$V amplitude, 180° phase separation |
| Strain signal sources | Two 20 Hz sinusoids, 1 mV amplitude, 180° phase separation |
| Behavioral noise | `10u*WHITE(...)` for the EMG sources; `100u*WHITE(...)` for the strain sources |
| Supply sources | Positive and negative rails defined with 30 V source values |
| Transient instruction | `.tran 0 40m 0` |
| Instrumentation-amplifier symbols | AD8429 |
| Additional amplifier symbols | OP07 |
| ADC / isolation symbols | AD4003 and ADuM4120A |

The saved LTspice EMG RC sections include $R = 40\,\Omega$, $C = 60\,\mu$F and $R = 60\,\Omega$, $C = 9\,\mu$F combinations. Their isolated first-order characteristic frequencies are approximately 66.3 Hz and 294.7 Hz, respectively. In the strain branch, 60 Ω with 400 $\mu$F gives approximately 6.63 Hz, and 80 Ω with 310 $\mu$F gives approximately 6.42 Hz. These values demonstrate why the saved

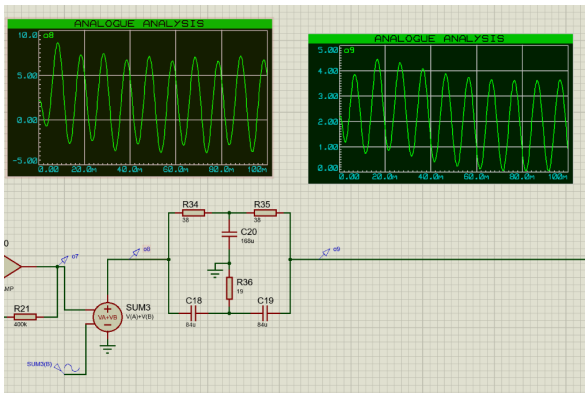
LTspice configuration cannot be described using the Proteus corner-frequency targets without qualification.

Two separate voltage sources are labeled as 50 Hz in `Project.asc`, but their saved sinusoidal amplitudes are **zero**. Consequently, that LTspice configuration does not, by itself, demonstrate rejection of injected 50 Hz interference. The 50-Hz rejection illustration in this report belongs to the Proteus design and its screenshots.

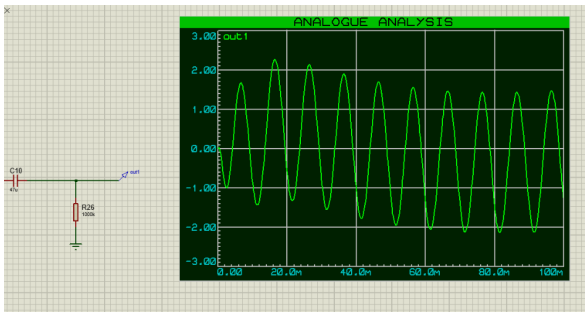
# 6. Simulation Results and Interpretation

## 6.1 EMG conditioning and power-line rejection

The Proteus waveforms illustrate the intended progression from a low-amplitude, interference-affected signal to a larger conditioned signal. The intermediate plots show the effect of the filtering stages and the amplification stages. The gains calculated above are consistent with the need to elevate microvolt-level EMG inputs to an observable output, although the supplied screenshots do not support a precise measured passband gain or a quantitative SNR-improvement claim.




(a) Notch stage and selected traces



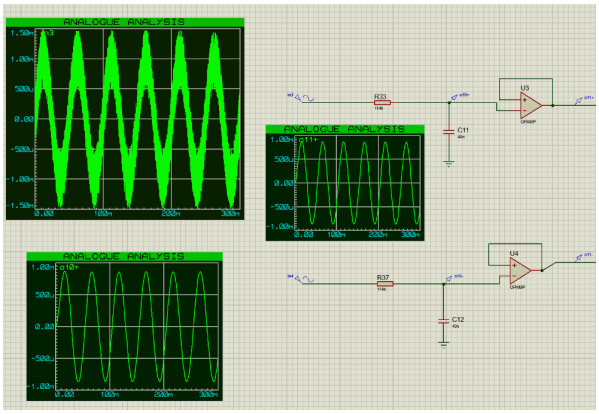

(b) Output coupling and waveform

**Figure 4:** Proteus EMG simulations: 50-Hz notch-filter demonstration and final DC-blocking stage. The displayed plots are the results supplied with the project.

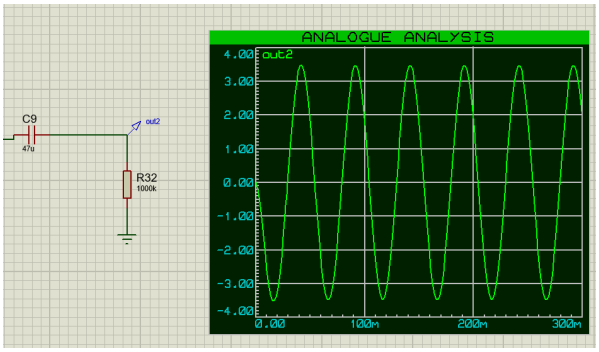
The displayed comparison at the notch stage is qualitative: the interfering component is visibly reduced in the plotted output. Numerical attenuation at 50 Hz, notch depth, bandwidth, and signal distortion were not extracted as frequency-domain measurements. The output coupling also produces a transient while its capacitor charges; this should be distinguished from steady-state sensor dynamics.

## 6.2 Strain-gauge conditioning

For the strain channel, the reported Proteus plots show a reduction of high-frequency fluctuations after low-pass filtering, followed by larger output amplitude after instrumentation and non-inverting amplification. The output coupling suppresses DC components, but it also attenuates signals sufficiently close to zero frequency, including very slow force variations.




(a) Low-pass filtering and buffering



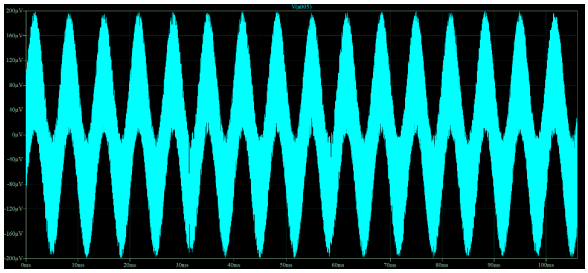

(b) Final DC-blocking output stage

**Figure 5:** Proteus strain-gauge channel, with the filtering and output conditioning presented in the original simulation results.

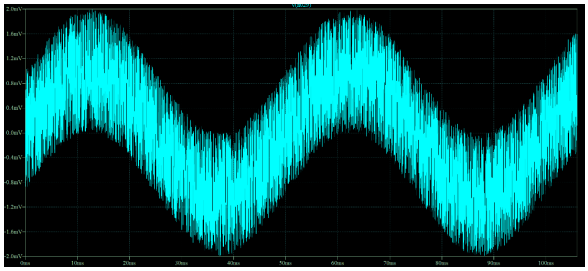
The traces demonstrate signal conditioning for the prescribed synthetic input. They do not provide force resolution, calibration accuracy, sensor hysteresis, or thermal-drift measurements. For an instrument intended to measure slowly varying grip force, the output high-pass cutoff and the low-pass bandwidth require selection against the actual motion time scale.

## 6.3 High-interference test cases

The supplied project also contains tests in which higher-amplitude interference was added to both simulated inputs. The figures below illustrate the change in waveform after passing through the processing chain. They are useful as visual examples of filtering but do not constitute an anomaly-classification algorithm.

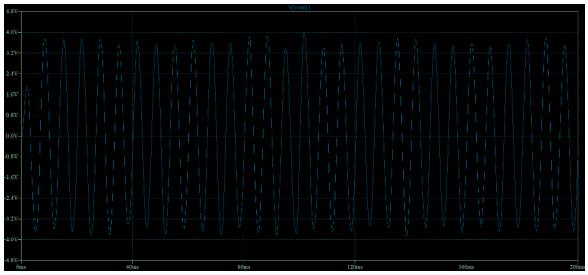


(a) EMG-like interference test

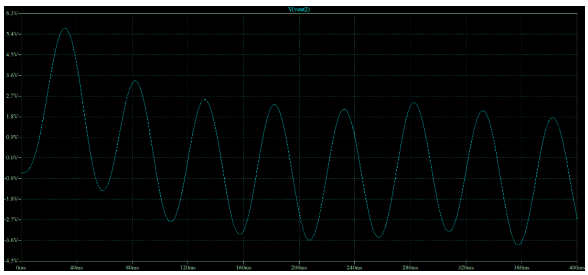


(b) Strain-like interference test

**Figure 6:** Input waveforms used for the high-interference examples. Both are synthetic test signals, not physiological or mechanical recordings.



(a) EMG channel output

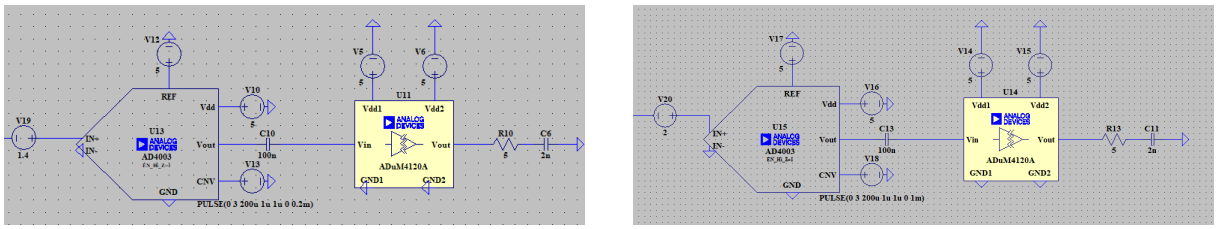


(b) Strain channel output

**Figure 7:** Outputs shown for the high-interference test cases. The waveform changes demonstrate filtering effects; no numerical noise-rejection ratio was provided.

### 6.4 ADC and isolation-stage exploration in LTspice

The additional LTspice section uses AD4003 converter symbols with a conversion-trigger signal and an ADuM4120A isolation-driver symbol. The converter input is shifted by a DC source in the schematic to keep the modeled voltage within the intended unipolar range. The source report includes plots labeled as quantized and scaled waveforms.




(a) EMG branch (b) Strain branch

**Figure 8:** ADC-related circuit sections included in the LTspice schematic. The isolation element is an ADuM4120A driver symbol; a complete microcontroller interface is not implemented.

These diagrams establish that ADC-related elements were included in the model. However, the source files do not include captured digital conversion codes, converter error analysis, an isolated digital interface, or a validated transfer of the converter output to a processor. In particular, a gate-driver isolation symbol is not sufficient evidence that the complete intended ADC interface provides galvanic isolation and correct digital data transport.

## 7. Design Assessment

**Table 3:** Consolidated evaluation of the circuit design and available evidence.

| Aspect | Assessment |
|---|---|
| Differential sensor interfaces | Implemented in the Proteus schematics and modeled with differential sources in LTspice. |
| EMG analog gain | Nominal two-stage design gain 6724 V/V; measured end-to-end gain is not separately documented. |
| Strain analog gain | Nominal two-stage design gain 220 V/V; force calibration is not documented. |
| Frequency selection | Proteus nominal EMG passband 20–200 Hz and strain low-pass near 35 Hz; these differ from the full input frequency ranges discussed in the introduction. |
| Mains interference | Notch behavior illustrated in Proteus; the saved LTspice 50 Hz test sources have zero amplitude. |
| Waveform conditioning | Qualitative filtering, amplification, and DC-blocking behavior is shown in supplied simulation screenshots. |
| ADC interface | Exploratory LTspice schematic provided; no complete validated digital acquisition or calibration result. |
| Physical validation | Not present in the provided material. No real EMG acquisition or measured force can be inferred. |

The principal design strength is the separation of two measurement channels with distinct gain and filtering choices. Buffer stages reduce interaction among the filters, and instrumentation amplification provides a reasonable architecture for small differential sensor outputs. The main limitations are the mismatch between some stated signal ranges and chosen cutoffs, the difference between the two simulators' component configurations, and the absence of quantitatively measured noise, bandwidth, gain, and ADC performance.

### 7.1 Bandwidth and amplitude considerations

For an ideal first-order low-pass section, the amplitude ratio at frequency $f$ is $|H(f)| = [1 + (f/f_c)^2]^{-1/2}$. With the stated strain-channel corner at 35 Hz, a 100 Hz component would have $|H| \approx 0.33$, or approximately $-9.6$ dB relative to the low-frequency passband. Thus, the selected cutoff emphasizes slow grip-force variations rather than retaining the entire 1–100 Hz range. This estimate applies to the ideal first-order section; it is not a measured overall circuit response.

The EMG amplification level likewise requires attention to signal headroom. At the nominal cascaded gain of 6724 V/V, an input differential component of 100 $\mu$V corresponds to approximately 0.67 V at the gain stages before filtering. The differential amplitude doubles if two electrode input sources each have that peak amplitude and are exactly 180° out of phase. Actual outputs are further affected by filter transfer functions and amplifier operating limits. These checks explain the need to specify whether a reported amplitude is per electrode or differential, and whether the measured node is before or after the filters.

The results should therefore be understood as an *analog front-end simulation study*. Converting the design into a measurement instrument would additionally require verified electrode/bridge interfaces, protection and bias networks, calibrated sensor excitation, appropriately scaled ADC input circuitry, and validation with actual sensor data.

## 8. Conclusion

A two-channel signal-conditioning system for triceps sEMG and grip-related strain-gauge signals was designed and examined in Proteus, with a supplementary LTspice model extending the concept toward digitization. The circuits combine frequency-selective filtering, buffering, instrumentation amplification, output gain, and offset removal. The submitted simulation figures illustrate filtering and voltage amplification of synthetic microvolt- and millivolt-level signals. Nominal gain calculations yield 6724 V/V for the EMG path and 220 V/V for the strain-gauge path before filter attenuation.

The evidence supports qualitative signal conditioning, but it does not establish quantitative sensing accuracy, actual muscle activation levels, calibrated grip force, or a fully validated digital acquisition interface. These boundaries are important when interpreting the circuits for a practical biomedical measurement application.